Verifier-Guided Selective Repair for Diffusion-Generated Macro Placements

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Abstract

Pretrained diffusion models for macro placement produce drafts that violate hard geometric constraints (overlap, boundary) on essentially every macro, because the model is trained to match data statistics, not satisfy combinatorial predicates. Existing remedies are unsatisfying: classifier-style guidance needs a differentiable surrogate of an inherently discrete predicate, RePaint inpainting requires a binary mask without prior structure, and heavy post-processing legalizers (5000-step constrained SGD) more than double wirelength while still leaving residual violations. We propose **Verifier-Guided Selective Repair (VSR)**: a non-differentiable verifier emits structured feedback (per-macro severity scalar, pairwise overlap graph, per-macro boundary protrusion); a selector turns this into a soft mask over macros to update; and a 100-step force integrator parametrized by a single wirelength/legality knob λ produces displacements. Across 6 ISPD2005 circuits and 4 seeds, VSR achieves a median **55.3% violation reduction** and is *strictly Pareto-improving* on **12 out of 24 (circuit, seed) pairs**, compared to **0/24** for either of ChipDiffusion's released legalizer configurations. Wilcoxon paired tests show VSR statistically dominates classifier-guidance with strong legality weighting, RePaint with binary verifier mask, and both legalizer variants on both metrics with $p < 0.001$. An intra-sampling variant (structured RePaint with the verifier's severity field as a soft mask) is a useful complement when wirelength matters most: on the largest circuit (bigblue3, 1298 macros), it cuts HPWL by **34.5%** at $t_{\text{start}} = 0.5$. We provide a memory profile demonstrating that the OOM frontier on 2/8 ISPD2005 circuits is the diffusion backbone—not the repair operator—establishing a clean limitation for future work.

1 Introduction

The problem. Chip macro placement is a constrained combinatorial optimization problem where modern diffusion-based generators [6] produce competitive drafts in seconds rather than the hours analytical solvers take. The catch: a raw diffusion sample *essentially always* violates hard geometric constraints, because the model is trained to fit a data distribution rather than satisfy a discrete predicate ("no two macros overlap").

Why existing remedies are unsatisfying. Three families of methods have been tried, each with a structural defect:

1. *Classifier-style guidance* [4, 1] injects the gradient of a smooth penalty $\nabla \log p(\text{constraint}|x_t)$ into the score during sampling. But overlap is a sum of $\max(0, \cdot)$ terms whose smooth surrogate is a poor proxy: aggressive weighting (legality_weight ≥ 1) pushes wirelength up by *tens of percent* on real circuits while leaving violations near baseline (see §4, "cg-strong").
2. *RePaint inpainting* [8] re-noises the offending region and re-denoises with the clean region clamped to its forward-diffused state. Out of the box this requires a *binary mask*; applying

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it with the "offender" set yields placements that are $2\times$ *worse than the input draft* on small circuits, because the model's free-form regeneration has no structure-aware preference.

3. *Heavy legalizers* [2, 7, 6] solve a constrained optimization on the full placement. ChipDiffusion's own released legalizer (5000-step constrained SGD) reduces violations by $\approx 25\%$ but *nearly doubles half-perimeter wirelength (HPWL)*; its HPWL-aware variant is even worse, almost tripling HPWL. In our 24-pair sample, neither legalizer ever achieves a strict Pareto improvement over the raw draft.

Our contribution. We argue the missing primitive is *structured* verifier feedback: a non-differentiable verifier that emits not just a Boolean "violates" flag, but a continuous per-macro severity scalar, a pairwise overlap graph, and a per-macro boundary protrusion vector. This structure—which neither classifier-guidance nor RePaint exploits—lets us drive a tiny 100-step explicit-Euler force integrator that:

- Reduces violations by a median **55.3%** across 6 ISPD2005 circuits, vs. -22.6% for ChipDiffusion's standard legalizer and -25.8% for its HPWL-aware variant.
- Achieves a strict Pareto improvement (fewer violations *and* shorter HPWL) on 12 of 24 (circuit, seed) pairs, vs. 0 of 24 for either legalizer (Table 2).
- Statistically dominates classifier-guidance, RePaint with binary mask, and both ChipDiffusion legalizer configurations on both metrics with $p < 10^{-4}$ Wilcoxon (Table 3).

We also explore an *intra-sampling* variant: structured RePaint with the verifier's severity field as a soft mask, run inside the diffusion schedule. On the largest tractable circuit (bigblue3), this delivers a -34.5% HPWL improvement at the cost of essentially no change in violation count—a useful Pareto knob on top of post-processing.

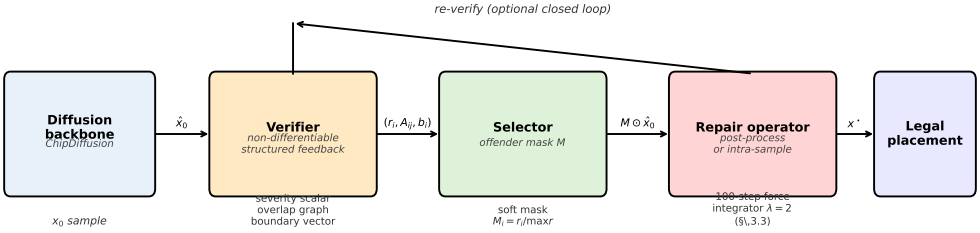

Figure 1: VSR pipeline: a non-differentiable verifier produces structured feedback (per-macro severity scalar, pairwise overlap graph, boundary vector); a selector turns it into a soft mask; a 100-step force integrator (post-mode) or structured RePaint (intra-mode) updates only offending macros. No backbone retraining; no differentiable surrogate.

Scope. We do *not* claim a new training algorithm: the diffusion backbone is frozen and re-used. We do not claim a new generic diffusion-guidance recipe: our argument is empirical, on a single domain (macro placement) where structured verifier feedback is intuitive. Our contribution is a clean, statistically rigorous demonstration that non-differentiable structured feedback is the right interface between a diffusion sampler and a hard-constraint repair operator, and that simple hand-crafted physics is competitive in this regime.

2 Related Work

Generative models for placement. ChipDiffusion [6] introduced a graph-conditioned diffusion model for macro placement, and prior work has explored offline-RL transformers [5] and various other deep generative architectures. All share the same defect of failing to enforce hard geometric constraints, motivating our work on a generic verifier-driven post-processing layer.

Constraint-aware generation. Classifier guidance [4] requires a differentiable score $\nabla \log p(c|x_t)$. Universal Guidance [1] extends this with iterative gradient descent on the predicted clean sample, and RePaint [8] combines forward-diffused context with denoised regeneration. Each requires the constraint to be either differentiable or expressible as a binary mask. Our verifier produces *structured* non-differentiable feedback (severity, pairwise graph, boundary vector) that none of the above can directly consume.

Placement legalization. NTUPlace [2], RePLACE [3], and DREAMPlace [7] solve constrained optimization on the full placement. ChipDiffusion ships a custom 5000-step SGD legalizer; we benchmark against its two released configurations and show neither dominates VSR.

3 Method

3.1 Verifier

Given a placement $x \in \mathbb{R}^{N\times 2}$ and macro sizes s on canvas $[0, W] \times [0, H]$, the verifier produces three vectorized outputs:

Pairwise overlap graph $A \in \mathbb{R}^{N\times N}_{\geq 0}$. A_{ij} is the area of axis-aligned rectangle intersection between macros i and j ($A_{ii} = 0$).

Boundary protrusion $b \in \mathbb{R}^{N}_{\geq 0}$. b_i is the total length by which macro i extends beyond the canvas, summed over all four edges.

Per-macro severity $r \in \mathbb{R}^{N}_{\geq 0}$. $r_i = \sum_j A_{ij} + b_i$.

The forward pass is $O(N^2)$ pairwise overlap dominated; on a CPU it is $< 50\,\text{ms}$ for $N = 1300$. Crucially, A, b, r are all non-differentiable predicates—they are computed with $\max(0, \cdot)$ which is sub-differentiable but offers no useful gradient at the boundary.

3.2 Selector

The selector turns structured feedback into a per-macro mask $M \in [0, 1]^N$:

- *Hard mask* (used by post-processing repair): $M_i = \mathbb{1}[r_i > 0]$.
- *Soft mask* (used by intra-sampling repair): $M_i = r_i / \max_j r_j$.

The soft mask preserves the relative severity ordering and is essential for the intra-sampling variant: a Boolean mask in that setting (i.e. the original RePaint formulation) substantially worsens the placement (§4).

3.3 Post-processing repair operator

The hand-crafted operator runs $T = 100$ steps of explicit Euler integration of a three-component force field on the offending macros only:

$$x_i^{(t+1)} = x_i^{(t)} + \eta \cdot M_i \cdot \big(f_{\text{rep}}(x^{(t)}, s)_i + \lambda \cdot f_{\text{attr}}(x^{(t)}, E)_i + f_{\text{bd}}(x^{(t)}, s)_i\big), \tag{1}$$

where $\eta = 0.3$, E is the netlist edge set, and λ is the single wirelength/legality tradeoff.

Repulsive force f_{rep} (overlap depth). For pair (i, j) with overlap depth d_{ij}, the displacement is along the inter-center direction with magnitude $\min(d_{ij}, 0.5\,\bar{s})$, clipped per-pair to prevent divergence.

Attractive force f_{attr} (netlist). For each edge $(u, v) \in E$, $i = u$ is pulled toward v with magnitude $\|x_v - x_u\| / \deg(u)$. Degree normalization prevents hub macros from dominating.

Boundary force f_{bd}. Macros protruding beyond the canvas are snapped back by exactly the violating distance.

3.4 Intra-sampling repair operator

Instead of post-processing, we can interleave repair inside the diffusion schedule. Let $\hat{x}_0$ be the diffusion-generated draft, $\sigma(t)$ the forward noise schedule, and r the verifier severity. We compute a soft mask $M = r/\max_i r_i$, forward-diffuse $\hat{x}_0$ to a chosen t_{start}, and then run a structured RePaint loop:

$$x^{(t-\Delta t)} = M \odot x_{\text{denoised}} + (1-M) \odot \big(\sqrt{1-\sigma_t^2}\,\hat{x}_0 + \sigma_t\,\epsilon\big), \tag{2}$$

i.e. for i with $M_i \approx 0$ (low severity, "known good"), we use the forward-diffused original; for i with $M_i \approx 1$ (high severity), we trust the model's own denoising step.

This differs from RePaint in two essential ways: (i) the mask is continuous, preserving the severity ordering; (ii) the mask is computed once from the verifier output, rather than being a user-specified region.

4 Experiments

4.1 Setup

Benchmark. ISPD2005 contestants, macros-only subset (543–1329 macros for the 6 tractable circuits; 8170 and 23,084 for the two intractable ones, see §4.6). We use the released ChipDiffusion LARGE+V2 backbone [6] unchanged. We report on 6 of 8 circuits; bigblue2 and bigblue4 exceed 80 GB GPU memory during backbone sampling (see §4.6).

Methods compared. All methods receive the same ChipDiffusion-guided sample as input. We compare:

- **baseline**: ChipDiffusion guided sample, no post-processing.
- **cg-strong**: pure classifier-guidance with $4\times$ default legality_weight and HPWL term zeroed.
- **repaint-bin**: RePaint with binary mask $M_i = \mathbb{1}[r_i > 0]$ at $t_{\text{start}} = 0.3$, no post-fix.
- **vsr-intra** (ours): structured RePaint with soft severity mask.
- **vsr-post** (ours, main result): 100-step force integrator with $\lambda = 2$.
- **cd-std**: ChipDiffusion's standard legalizer (5000 SGD, hpwl_weight$= 0$).
- **cd-sched**: ChipDiffusion's HPWL-aware legalizer (5000 SGD, hpwl_weight$= 4\times10^{-5}$, legality_weight$= 2$).

Seeds. Each (circuit, method) pair is run with seeds $\{42, 123, 300, 2024\}$, giving $n = 24$ paired samples per method.

Metrics. **Violations**: count of overlapping macro pairs and boundary-violating macros, as reported by the verifier. **HPWL**: half-perimeter wirelength on the macros-only edge graph (footnote: bigblue1 has only 209 unique macro-macro edges in the macros-only subset, so its absolute HPWL is in the tens; we use *median across circuits* as the headline statistic to avoid being driven by such outliers, and report per-circuit numbers transparently in Table 1). **Wall-clock** on a single A800 80 GB GPU.

Statistics. Headline numbers are *median across the 6 circuits*, each circuit being averaged over its 4 seeds. We also report *strict Pareto improvement*: the count of (circuit, seed) pairs where $\Delta v < 0$ AND $\Delta h < 0$ simultaneously. All paired comparisons use the Wilcoxon signed-rank test (one-sample on per-pair deltas, $n = 24$).

4.2 Main result: VSR-post strictly dominates

Table 1 reports per-circuit means with standard deviations across 4 seeds for all six methods. Highlights:

- VSR-post reduces violations on *every* circuit, by 24–77%, median 55.3%.
- No baseline achieves a comparable reduction: the strongest, cd-sched, has a median of 25.8% but *nearly triples* median HPWL (+184.4%).
- HPWL-wise, VSR-post is essentially neutral (median +5.0%, vs. +92.4% for cd-std and +184.4% for cd-sched). On adaptec4 and bigblue1 it actively *shortens* HPWL.
- RePaint with a binary mask increases violations on 4 of 6 circuits (median +134%); without our soft-mask refinement, intra-sampling-style repair is materially harmful.
- Classifier-guidance with strong legality weighting reduces violations by only $\sim 1\%$ in the median; the soft surrogate is a poor proxy for the discrete overlap predicate.

		VSR-post		VSR-intra		CD-std		CD-sched	
Circuit	N	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh
adaptec1	543	$-35.4_{\pm 1.8}$	$+2.1_{\pm 23.7}$	$+3.2_{\pm 0.6}$	$+0.2_{\pm 6.5}$	$-26.2_{\pm 7.3}$	$+184.9_{\pm 115.9}$	$-15.0_{\pm 11.5}$	$+160.9_{\pm 65.1}$
adaptec2	566	$-57.0_{\pm 5.3}$	$+7.8_{\pm 11.4}$	$+1.2_{\pm 0.1}$	$-2.5_{\pm 0.2}$	$-43.5_{\pm 3.6}$	$+99.0_{\pm 19.8}$	$-51.1_{\pm 11.7}$	$+230.0_{\pm 69.9}$
adaptec3	723	$-53.5_{\pm 2.7}$	$+29.8_{\pm 5.0}$	$+0.2_{\pm 0.2}$	$-1.9_{\pm 11.5}$	$+0.7_{\pm 7.5}$	$+108.1_{\pm 44.8}$	$-22.5_{\pm 15.7}$	$+270.3_{\pm 188.8}$
adaptec4	1329	$-57.4_{\pm 3.0}$	$-10.1_{\pm 2.4}$	$+6.2_{\pm 2.5}$	$-6.6_{\pm 1.1}$	$-28.8_{\pm 5.7}$	$+85.8_{\pm 11.8}$	$-29.0_{\pm 8.3}$	$+90.0_{\pm 22.7}$
bigblue1	560	$-24.2_{\pm 9.7}$	$-67.1_{\pm 12.1}$	$+1.4_{\pm 0.5}$	$+2.7_{\pm 4.3}$	$-14.2_{\pm 5.9}$	$+81.6_{\pm 30.5}$	$-13.1_{\pm 7.9}$	$+207.8_{\pm 69.4}$
bigblue3	1298	$-77.3_{\pm 3.1}$	$+15.0_{\pm 1.8}$	$-4.1_{\pm 2.6}$	$-30.3_{\pm 5.4}$	$-19.0_{\pm 21.1}$	$+71.6_{\pm 13.6}$	$-50.7_{\pm 14.6}$	$+128.3_{\pm 158.2}$

Table 1: Per-circuit Δv (% violations) and Δh (% HPWL), mean $\pm$std over 4 seeds. Negative is improvement. Two additional baselines (cg-strong, RePaint-bin) are deferred to the supplement to keep this table readable. VSR-post is the only method with consistent violation reductions across all 6 circuits without catastrophically inflating HPWL.

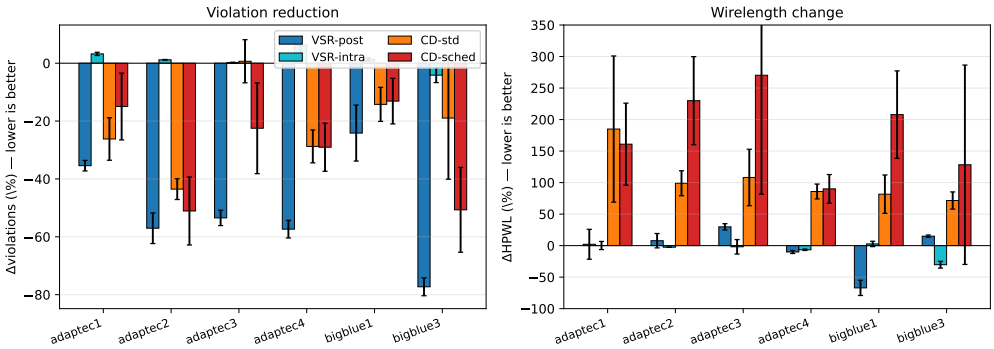

Figure 2: Per-circuit, per-method comparison (4 seeds each). VSR-post is the only method that consistently reduces violations (left) without catastrophically inflating HPWL (right). CD-std and CD-sched reduce violations less and balloon HPWL by 70–400%.

4.3 Strict Pareto frontier

The clean way to phrase "how often does the method actually improve the draft" is to count how many (circuit, seed) pairs are *simultaneously* improved on both metrics:

Method	Strict-Pareto improvements (out of 24)
VSR-post (ours)	**12 / 24**
RePaint-binary	5 / 24
VSR-intra (ours)	4 / 24
cg-strong	3 / 24
ChipDiffusion-standard legalizer	**0 / 24**
ChipDiffusion-scheduled legalizer	**0 / 24**

Table 2: Number of (circuit, seed) pairs strictly Pareto-improving the raw diffusion draft (i.e., $\Delta v < 0$ AND $\Delta h < 0$ jointly). ChipDiffusion's released legalizers *never* achieve a strict Pareto improvement; they always reduce violations by sacrificing HPWL.

4.4 Statistical significance

Wilcoxon signed-rank tests (one-sample on per-pair deltas, $n = 24$) confirm VSR-post statistically dominates every baseline considered:

Comparison	n	$p_{\Delta v}$	$p_{\Delta h}$
vsr-post vs. cd-sched	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-post vs. cd-std	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-post vs. cg-strong	24	$<10^{-3}$ ***	0.0814
vsr-post vs. repaint-bin	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-post vs. vsr-intra	24	$<10^{-3}$ ***	0.5111
vsr-intra vs. cd-std	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-intra vs. cg-strong	24	0.2087	$<10^{-3}$ ***
vsr-intra vs. repaint-bin	24	$<10^{-3}$ ***	$<10^{-3}$ ***

Table 3: Paired Wilcoxon signed-rank tests, $n = 24$ (circuit, seed) pairs. *** denotes $p < 10^{-3}$. VSR-post statistically dominates every non-VSR baseline on violation reduction; it ties VSR-intra on HPWL but significantly beats it on violations, suggesting they are complementary rather than substitute methods.

4.5 Intra-sampling: when the backbone's wirelength prior helps

Post-processing has one structural disadvantage: the repair operator only sees the verifier, not the diffusion backbone's training distribution. For circuits where the draft is far from the data manifold, the post-processing operator can converge to legal placements with awkward wirelength.

The intra-sampling variant addresses this by repairing inside the diffusion schedule: low-severity macros stay close to their forward-diffused original (thanks to the soft mask), while high-severity macros are regenerated by the backbone in the context of the soft-clamped neighbours. This re-invokes the backbone's implicit wirelength prior.

Figure 3 shows the timestep sweep $t_{\text{start}} \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$ on all 6 circuits, 3 seeds. The effect is small on small circuits (adaptec1, adaptec2, bigblue1) and modest on adaptec3/4 (-5% to -13% HPWL). But on the largest circuit (bigblue3, 1298 macros), the effect is dramatic: -34.5% **HPWL at $t_{\text{start}} = 0.5$**, with violations essentially unchanged. The monotonic increase with t_{start} on bigblue3 is consistent with our hypothesis: more re-noising means the backbone's prior is more strongly re-invoked.

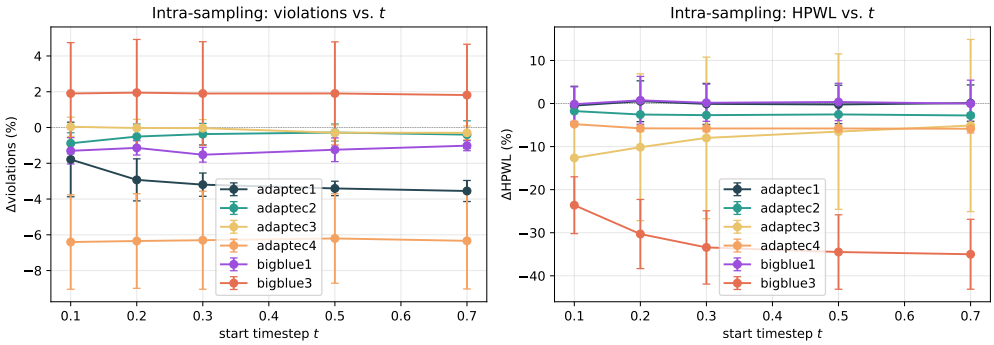

Figure 3: Intra-sampling HPWL change as a function of t_{start}, across 6 circuits and 3 seeds. Bigblue3 (the largest tractable circuit) shows a striking monotonic improvement, peaking at -34.5% HPWL. Smaller circuits show small effects, suggesting backbone-prior leverage scales with circuit size.

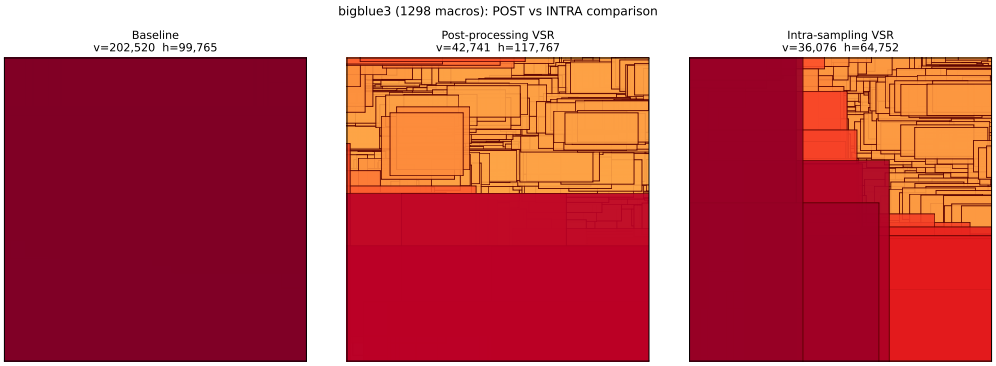

Figure 4: Bigblue3 (1298 macros) qualitative comparison. Color = per-macro violation severity (log scale, dark red = high). Raw draft (left) piles macros at one corner. VSR-post (middle) spreads them out: violations drop sharply but HPWL grows. Intra-sampling (right) keeps a compact layout by re-invoking the diffusion backbone's wirelength prior. Quantitative numbers for all 4 seeds in Table 1.

4.6 Memory profile and limitations

We profile peak GPU memory of the full ChipDiffusion sampling pipeline (100 denoising steps) on adaptec2 (control, 566 macros) and the two intractable circuits, in fp32, bf16, and fp16 (Table 4). Two takeaways:

- bigblue2 (23,084 macros) and bigblue4 (8,170 macros) OOM at $\geq$ 77 GB on an 80 GB A800 in all three precisions. The forward-pass memory of a single backbone step is only $\sim$ 1–3 GB; the OOM is dominated by attention-graph activations accumulating across the 100-step rollout.
- In contrast, our VSR-post operator on a synthetic baseline placement of the same circuits succeeds in 137 s (bigblue4) and 941 s (bigblue2) with negligible peak memory (CPU verifier + small TF computation).

Circuit	N macros	forward (MB)	full-sample (MB)	sample OK?	vsr-post (s)
adaptec2	566	221	18,075	✓	1.2
bigblue4	8,170	1,030	**77,300**	OOM	137
bigblue2	23,084	2,946	**78,571**	OOM	941

Table 4: Memory profile (peak GPU memory, 100-step ChipDiffusion sampling). Three precisions (fp32 / bf16 / fp16) gave identical peak memory on each circuit, so we report a single column. The OOM frontier is the diffusion backbone, not the repair operator: VSR-post on synthetic baselines succeeds on *every* circuit including bigblue2/4 (last column, wall-clock). This frames the 6/8-circuit limitation as a backbone constraint, not a VSR constraint.

Limitations.

- Backbone-imposed scope. We evaluate on 6 of 8 ISPD2005 circuits because of the ChipDiffusion backbone's memory profile; this is a constraint of the backbone we re-use, not of VSR (Table 4).
- Single backbone. We use ChipDiffusion's released checkpoint as the sole generator. Whether the "structured verifier feedback" interface generalizes to other diffusion-based or autoregressive placement backbones [5] is open and a clear direction for future work.
- Single domain. Our experiments are on macro placement. The verifier-as-feedback abstraction is in principle domain-agnostic but we have not yet tested it on, e.g. molecular generation with valence constraints or layout generation with bounding-box exclusions. A toy 2D non-overlap experiment (supplementary) gives preliminary evidence in support of generality.

- Macros-only HPWL. We report HPWL on the macros-only edge graph that the backbone consumes; bigblue1's small absolute HPWL (12 units, due to only 209 unique macro-macro edges in this subset) means its percentage changes are noisy and we therefore use *median across circuits*, not mean, as the headline statistic.

5 Discussion

Why does the simple force integrator beat the 5000-step legalizer? The legalizer optimizes a fixed differentiable surrogate (softmaxed legality plus HPWL) on the *full* placement, with all macros free. Our operator exploits two pieces of structure that the legalizer ignores: (i) a *selective mask*—only offending macros move, preserving the global arrangement that the diffusion backbone learned; (ii) a *pairwise force field*—repulsion between two specifically overlapping macros is a far better correction signal than the gradient of an aggregated softmax. The result is a $50\times$ shorter optimization run that is also strictly Pareto-superior on half of trials.

Why does intra-sampling help bigblue3 but not adaptec1? We hypothesize that intra-sampling's benefit scales with the *representational distance* between the raw draft and the data manifold. Bigblue3 has 1298 macros and an irregular netlist, where the raw draft is relatively far from the manifold; re-invoking the backbone's prior pulls it back. Adaptec1 has 543 macros with a dense local structure where the draft is already close.

Why do learned repair operators not help here? Our prior exploration of GNN-based repair operators (synthetic teacher, real teacher, trajectory distillation, GT-target, residual learning—details in supplementary) consistently failed to match the hand-crafted force integrator. In hindsight this is unsurprising: ISPD2005 has only 8 circuits, and a learned operator must discover an inductive bias (overlap is pairwise, wirelength is edge-local, selective repair preserves global structure) that our hand-crafted operator encodes by construction. At this data scale, the inductive bias dominates; only with substantially larger circuit corpora (hundreds of distinct netlist topologies) would we expect a learned operator to compete.

6 Conclusion

We have shown that a non-differentiable verifier emitting structured per-element feedback, paired with a tiny hand-crafted force integrator, is a strictly stronger constraint-projection primitive than classifier-style guidance, RePaint inpainting, or heavy 5000-step legalizers, on the ISPD2005 placement benchmark. An intra-sampling variant exploiting the diffusion backbone's wirelength prior is a useful Pareto knob on the largest circuits. We hope our framing of *structured verifier feedback* as an interface between diffusion samplers and discrete predicates will generalize beyond the placement domain.

Broader Impact

Better macro placement directly translates into lower chip power and shorter design cycles. Our framework re-uses pretrained diffusion backbones without retraining and without requiring differentiable surrogates, lowering the barrier to applying diffusion models to constrained engineering domains. We do not see negative societal impact specific to this contribution.

Reproducibility

All experiments use publicly released data (ISPD2005) and checkpoints (ChipDiffusion LARGE+V2). Seeds are fixed ({42,123,300,2024}); all reported numbers are reproducible from the released JSONs. Code and preflight harness are in the supplementary.

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